

Probability problems

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1 Counting and naive probability

1.1 Basic count I

A restaurant orders Apples, Bananas, Oranges, Strawberries, Grapes, Pears, and Watermelon. Its chef wants to make a fruit salad.

1. If a salad must contain exactly 3 different kinds of fruit, then how many different fruit salads can the chef make?
2. How many different fruit salads can the chef make without the above restriction? A salad must contain at least one ingredient.
3. Suppose the chef never mixes bananas and apples together. How many different fruit salads with exactly three ingredients can she make given this restriction?
4. The chef decides to make a random salad with exactly three ingredients. What is the probability that the salad does not have at least one of the following ingredients: banana, apple, orange.

1.2 Blood types

		Donor's blood type							
		O-	O+	B-	B+	A-	A+	AB-	AB+
Recipient's blood type	AB+	✓	✓	✓	✓	✓	✓	✓	✓
	AB-	✓		✓		✓		✓	
	A+	✓	✓			✓	✓		
	A-	✓				✓			
	B+	✓	✓	✓	✓				
	B-	✓		✓					
	O+	✓	✓						
	O-	✓							

There are 8 blood types. A living kidney transplant can only occur between a donor-recipient pair with compatible blood types, as indicated in the table above. For example, a donor with type A^+ is compatible only with recipients with A^+ and AB^+ blood types.

Imagine a couple where one partner needs a transplant (the recipient) and the other is willing to donate a kidney (the donor). The problem is that the blood types of the two people in a couple are not always compatible. Assume that blood types in a couple are independently and discretely uniformly distributed — each of the 8 blood types is equally likely.

1. How many possible combinations of donor-recipient blood types are there?
2. How many compatible combinations of donor-recipient blood types are there?
3. What is the probability that a couple has compatible blood types?
4. What is the probability that the donor and the recipient both have blood type B⁺?
5. What is the probability that two people in a couple have different blood types?
6. Out of 100 donor-recipient couples, on average, how many have compatible blood types?
7. A transplant center has 5 donor-recipient couples. What is the probability that none of the couples have compatible blood types?
8. Donor Alice and recipient Eve have compatible blood types. What is the probability that Alice has blood type O⁻?
9. In reality, blood types are not equally likely. Suppose the probability of blood type B⁺ is 0.2. What is the smallest number of people we need to test to have greater than 0.5 probability of finding at least one person with this blood type?

1.3 McDonald lottery

McDonald's in Mega Silk Way ran a lottery. Each customer received a scratch ticket that had 11 covered spots. Behind the spots there were 3 burger, 3 Cola, 3 fries, and 2 McDonald's symbols. Here is an example of a ticket with all the spots scratched off.



1. How many ways are there to scratch off a single spot such that there is a Cola symbol behind it?
2. How many ways are there to scratch off two spots such that there is a Cola symbol behind each scratched spot?
3. How many ways are there to scratch off three spots such that there is a Cola symbol behind each scratched spot?
4. How many ways are there to scratch off two spots such that there is no Cola behind each scratched spot?
5. How many ways are there to scratch off three spots such that there is no Cola and no Fries behind each scratched spot?

6. You randomly choose a single spot to scratch off. What is the probability that there is a Cola symbol behind it?
7. You randomly choose three spots to scratch off. What is the probability that there is a Cola symbol behind each of them?
8. You randomly choose three spots to scratch off. Assume that you win, if the three symbols behind the scratched off spots coincide. What is the probability to win? You can leave your answer as a fraction of binomial coefficients.
9. You randomly choose three spots to scratch off. Now assume that McDonald's symbol can be used to substitute any other symbol (like a joker in poker). What is the probability to win? You can leave your answer as a fraction of binomial coefficients.

1.4 Telegram chats

I chat with five friends on Telegram. For every nonempty subset of these friends—including individual friends—I have created a separate group chat.

1. How many chats include exactly 3 friends?
2. How many different chats do I have?
3. How many group chats include at least 2 friends?
4. Oleg is one of my friends. How many chats are there with Oleg?
5. Oleg does not like my other friend Kamila. Therefore I have deleted all chats which have them both. How many chats do I have left?
6. From now on, suppose that I have only chats that do not have both, Oleg and Kamila, at the same time. I receive some news. Let A_i be the event that my friend i needs to know this news. Assume $A_1, \dots, A_5$ are independent, and $\mathbb{P}(A_i) = p$ for all i . Let X be a random variable equal to the number of friends who need to know the news. Find the PMF of X .
7. Continuing from (d), let Y be a random variable equal to the smallest number of messages I have to send in order to tell the news to everyone who needs to know it. Find the PMF of Y .
8. Find expectation of Y .
9. How many pairs of group chats are such that one group is a subset of the other?

1.5 Random quiz

A university course consists of 20 lectures and must have exactly two quizzes. Each quiz is conducted at the end of a lecture and covers the material in the lecture. A professor chooses two lectures that will have quizzes. Provide an arithmetic expression for each answer.

1. How many choices does the professor have?
2. How many choices are there such that there is no quiz during the first **and** the final lecture?

3. How many choices are there such that there is no quiz during the first **or** the final lecture?
4. How many choices are there such that there is at least one lecture without a quiz in between the two lectures with a quiz?
5. Suppose the professor randomly chooses the two lectures that will have a quiz. Alice got sick and has to miss the first two lectures. What is the probability that she will miss at least one quiz?
6. Suppose the professor randomly chooses the two lectures that will have a quiz. Alice missed the first two lectures and it turned out that none of them had a quiz. Now Alice is considering skipping a third lecture too. Given this, what is the probability that the third lecture won't have a quiz?

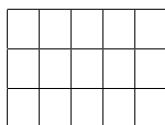
1.6 Balls and boxes

There are 5 distinguishable balls and 6 distinguishable boxes. Provide an arithmetic expression for each answer.

1. What is the number of ways to put each of 5 balls into one of 6 boxes?
2. What is the number of ways to put each of 5 balls into one of 6 boxes such that none of the 6 boxes has more than a single ball?

1.7 Random cars

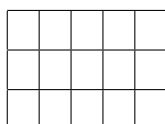
Consider a rectangular road network below. Provide an arithmetic expression for each answer.



1. Suppose a car needs to travel from the lower left corner to the upper right corner. How many different paths the car can take if it can only move up or to the right (never down or left)?
2. Suppose that each time when facing an intersection the car randomly chooses one of the allowed turns. For example, at the start, it chooses to go up or right with equal probabilities. For an intersection, find the probability that the car will pass the intersection?
3. Suppose Car 1 starts in the lower left corner and moves only up or right, while car 2 starts at the same time at the upper right corner and moves at the same speed only down or left. At each junction each car independently takes one of the possible directions with equal probability. What is the probability that the cars will meet? You can give the answer in terms of the sum of terms.

1.8 Counting rectangles

Consider the figure below.



1. How many rectangles are there in the figure (a rectangle can consist of several smaller rectangles)?
2. Generalize your result for any $m \times n$ initial grid (for example the figure above contains 6x4 grid).
3. Now consider removing $k \times l$ rectangle in the upper right corner of the initial rectangle, where $k < m$ and $l < n$. How many rectangles are there in such an incomplete rectangle?

1.9 Papers in a hat

Alice rolls three dice and records the result of each roll on separate slips of paper, which she puts into a hat.

1. What is the probability that all three dice rolls were a 6?
2. You draw two of the slips, each showing a 6. What is the probability that all three dice rolls were a 6?
3. Alice tells you that the sum of the three rolls is greater than or equal to 15. What is the probability that all rolls were a 6?

1.10 Round-robin tournament

A round-robin tournament is being held with 6 tennis players; this means that every player will play against every other player exactly once. Provide an arithmetic expression for each answer.

1. How many games does the tournament have?
2. Alice is one of the 5 players. How many possible outcomes are there for the tournament (an outcome specifies who won and who lost each game) such that Alice wins all games that she plays?

1.10.1 PIN codes

A phone PIN consists of 4 digits (each digit is chosen from the 10 available).

1. How many different PIN codes are possible?
2. Assuming the PIN is chosen uniformly at random, what is the probability of guessing the PIN on the first attempt?
3. Assuming the PIN is chosen uniformly at random, what is the probability of guessing the PIN within the first three attempts?
4. You are told that the PIN contains the digits 1, 2, 3, and 4, but you do not know the order. How many possible PIN codes are consistent with this information?
5. Now suppose the PIN uses the digits 1, 2, and 3, and *one* of these digits is repeated twice—but you do not know which digit is repeated. How many different PIN codes satisfy this description?
6. A PIN is chosen uniformly at random. Let the random variable X be the number of times the digit “6” appears in the PIN.

- Is X a continuous or a discrete random variable?
- What is the probability that $X = 0$?
- Find $\mathbb{E}[X]$.

1.11 Miscellaneous problems

1. How many ways are there to permute the letters in the word BOSS?
2. A salesman must visit each of the 4 cities. If he visits each city exactly once, how many routes are there that the salesman can take (note, a route is the order in which the salesman visits the cities)?
3. There are 5 economics and 6 math course on offer. A student must take 2 courses from each field. How many different combinations of courses are there to choose from?
4. How many ways are there to split 10 people into 2 teams, where one team has 6 people, and the other team has 4 people?
5. How many ways are there to split 20 people in two teams of 10 people each?
6. There are 6 boys and 4 girls. How many ways are there to form a team of 3 people, including a captain, when a captain must be a girl?
7. You are randomly given 5 cards out of a standard deck with 52 cards. What is the probability to have all 4 aces?
8. Are events A and A^c independent (it is enough to say yes or no)?
9. Two dice are thrown. What is the expected sum of the scores on two dice?
10. How many functions are there from a set A with 10 elements to the set of all subsets of the set A (including the set itself and the empty set)?
11. Each of n balls is independently placed into one of n boxes, with all boxes equally likely. What is the probability that exactly one box is empty?
12. In a shop, there are 10 kinds of postcards. We want to send postcards to 5 friends. Suppose we want to send two different cards to each of them, and no two people can receive the same kind of postcard. In how many different ways can this be done?
13. Assel, Bota, and Chingiz all must take ECON211. There are two sections in the class and students are randomly assigned to sections. What is the probability that they are in the same section?
14. Tyrion, Cersei, and ten other people are sitting at a round table, with their seating arrangement having been randomly assigned. What is the probability that Tyrion and Cersei are sitting next to each other?
15. How many different ordered triplets of non-negative integers a, b , and c are there such that $a + b + c = 2020$? Leave your answer as a binomial coefficient.
16. Alice and Assel each randomly independently choose k out of n classes, where $k \leq n$. Let X be the number of classes taken by both students at the same time. Find $\mathbb{E}(X)$ (in terms of k and n).

17. Three friends, Asel, Beibit, and Chingiz were admitted to NU and will get a place in a dorm. The following are the numbers of available rooms in each of the dorm blocks. Block 22 has 5 rooms; block 23 has 5 rooms; block 24 has 10 rooms. Moreover, each available room has two vacant places. New students are randomly assigned a place in a dorm (each assignment of places is equally likely). What is the probability that all three friends will live in the same block?
18. What is the probability that at least one 6 appears when 6 fair dice are rolled?

2 Non-naive definition and Conditional probability

2.1 Basic probabilities I

The weather forecast for the next month specifies the following probabilities of rain for the next four Sundays: 0.4, 0.5, 0.1, and 0.8. Assume that the rains are independent from each other.

1. What is the probability that it rains on the first and second Sunday?
2. What is the probability that it rains at least once during the four Sundays?
3. What is the probability that it rains exactly once during the four Sundays?

2.2 Disease testing

A new treatment for a disease is being tested, to see whether it is better than the standard treatment. The existing treatment is effective on 50% of patients. It is believed initially that there is a $\frac{2}{3}$ chance that the new treatment is effective on 60% of patients, and a $\frac{1}{3}$ chance that the new treatment is effective on 50% of patients. In a pilot study, the new treatment is given to 20 random patients, and is effective for 15 of them. Given this information, what is the probability that the new treatment is better than the standard treatment?

2.3 Coin flips

A coin is tossed three times. What is the probability that exactly two heads occur, given that:

1. the first outcome was a head?
2. the first outcome was a tail?
3. the first two outcomes were heads?
4. the first two outcomes were tails?
5. the first outcome was a head and the third outcome was a head?

2.4 Maja's and Oleg's game

Maja and Oleg play a game. Initially they start with some quantity of soldiers each. Each round they have a battle where a probability to win a battle for a player with k soldiers against an opponent with l soldiers is $\frac{k}{k+l}$. If a player wins a battle, then he or she takes one soldier from the opponent. A player who runs out of soldiers loses (and the other one wins). Find a probability that Maja wins the game when she initially has 2 soldiers and Oleg has only 1.

2.5 Prisoner puzzle

A prisoner is given a chance for parole. He's blindfolded and told to choose one of two bags; once he does, he is to reach in and pull out a marble. Each bag has 25 red and 25 blue marbles, and the marbles all feel the same. If he pulls out a red marble he is set free; if it's a black, his parole is denied.

1. What is his chance of winning parole?
2. Now the prisoner is free to distribute the marbles among the two bags however he wishes, so long as all the marbles are distributed. After he distributes the marbles, he is blindfolded, and chooses a bag at random and then a marble. Find the highest probability of being set free the prisoner can achieve.

2.6 Ameoba problem

A single amoeba, Bobo, lives in a pond. After one minute Bobo will either die with probability $2/3$ or split into two amoebas with probability $1/3$, and in subsequent minutes all living amoebas will behave the same way, independently. Provide an arithmetic expression for each answer.

1. What is the probability that some amoebas will be alive after 2 minutes?
2. What is the probability that the amoeba population will eventually die out?

2.7 Balls and boxes

There are two boxes with balls. Box 1 has 10 black and 15 white balls; Box 2 has 10 white balls only. A box is chosen at random with equal probability. You do not know which box is chosen and randomly take two balls out of the chosen box without replacement.

1. Is the color of the first ball you take independent from the color of the second ball?
2. Given that both chosen balls are white, what is the probability that Box 1 is chosen?
3. What is the probability that the chosen balls are of the same color?

2.8 Two-man rule

During the Cold War, the two-man rule was used to prevent accidental launch of nuclear weapons. For example, on U.S. bombers, two operators separately receive the authorization code for launching the nuclear weapon. Operators compare their codes and proceed with the launch if and only if their codes are exactly the same.

Suppose a code consists of a string of zeros and ones of length n . Let E_{jk} denote the event that operator $j = 1, 2$ makes a mistake when recording digit k , where $k = 1, 2, \dots, n$. A mistake means writing a 0 instead of 1 or vice versa. Assume that all events E_{jk} are independent, and $P(E_{jk}) = \varepsilon$ for all j, k . Let $n = 4$ and $\varepsilon = 0.1$.

1. What is the probability that operator j makes **at least one** mistake in his code?
2. What is the probability that operator j makes **exactly one** mistake in his code?
3. What is the expected number of mistakes operator j makes?
4. What is the probability that after comparing the codes, the operators agree on the **correct** code?

2.9 Fire alarm

If there is a fire, a fire alarm rings with probability $q \in (0, 1)$. If there is no fire, then sometimes a fire alarm still rings. This happens with probability $p \in (0, 1)$. The prior probability of fire is 0.01. Provide an algebraic expression for each answer.

1. What is the probability that the alarm rings?
2. What is the probability that there is a fire, given that you hear a fire alarm?
3. Let A be an event that the alarm rings and F be an event that there is a fire. Are A and F disjoint?
4. Explain what event $A \cap F$ is in terms of the conditions of the problem.
5. Explain what event $A \cup F$ is in terms of the conditions of the problem.
6. Explain
7. Find $P(A^c \cap F^c)$?
8. Find $P(A^c \cup F^c)$?
9. Find condition on q and p such that A and F are independent.

2.10 Sentence Generation by a Language Model

A simplified language model generates sentences one word at a time, beginning with the special token **START**, and ending with **END**. At each step, the model samples the next word based on the current word, according to the following transition probabilities:

From $\rightarrow$ To	cat	runs	fast	END
START	0.4	0.5	0.1	0.0
cat	0.2	0.5	0.0	0.3
runs	0.1	0.2	0.4	0.3
fast	0.0	0.0	0.0	1.0

For example, if the current word is "cat", then given this the conditional probability of the next word being "runs" is 0.5.

Answer the following questions:

- (a) What is the probability that the model generates the sentence:
START $\rightarrow$ **runs** $\rightarrow$ **fast** $\rightarrow$ **END**?
- (b) What is the expected number of words (not counting **START**, but counting **END**) in a sentence generated by this model?
- (c) What is the probability that a generated sentence contains the word **cat** at least once?

2.11 Miscellaneous problems

- An exam has 3 true and false questions. Each true and false question has two answer options, and only one of the options is correct. Abu is a monkey who takes the exam. He randomly picks an answer to each question. What is the probability that Abu makes at least one mistake?
- Two dice are thrown. Find the expectation of the sum of the scores if it is known that different sides came up.
- There are 20 boys and 10 girls taking the first section of ECON211, and 30 girls and 20 boys taking the second section of ECON211. No student takes both sections. A student is chosen at random out of the students taking ECON211. The chosen student is a girl. What is the probability that the chosen student takes the first section?
- Studies suggest that the chance of conceiving for an average couple trying to have a child are 15% per month. Assume that conceiving has the same probability each month and is independent across months. Find the probability that a couple will not conceive after a year of trying.
- A family has 3 children, named A, B, and C. Find the probability that A is older than B, given that A is older than C.

3 Discrete random variables and their distributions

3.1 Party guests birthdays

People are arriving at a party one at a time. While waiting for more people to arrive they entertain themselves by comparing their birthdays. Let X be the number of people needed to obtain a birthday match, i.e., before person X arrives there are no two people with the same birthday, but when person X arrives there is a match. Find the PMF of X .

3.2 Sequence of investments

You have worked for years as a management consultant and saved an amount $I > 0$. You finally decide to quit and work for yourself. Over these years, you came up with N investment ideas (let's call them projects), labeled by $k = 1, 2, \dots, N$. To implement project k , you must invest exactly your entire wealth I . Then:

- With probability $P_k \in (0, 1)$, you get a return $R_k > 0$ on your investment,
- With probability $1 - P_k$, your investment is completely lost.

Assume $\sum_{k=1}^N P_k R_k < 1$. These conditions imply that projects can only be implemented one at a time. A successful return means you receive $I(1 + R_k)$ after investing I .

You cannot borrow money and aim to maximize your expected wealth. Assume $N = 2$.

1. Which project(s) do you want to implement? Derive the condition on the parameters.
2. Characterize the optimal investment rule as a function of parameters. The rule must specify:
 - Which project(s) you choose to implement,

- In which sequence you implement them, if more than one.

Explain clearly how you derive the rule.

3. Interpret your rule in terms of *odds of success* of the projects (see Stat 110 Homework 3, Fall 2011, for the definition of odds).
4. Compute your maximum expected wealth as a function of the parameters.
5. **(Extra)** Generalize the solution for arbitrary $N > 2$. Prove your conjecture as rigorously as possible.

3.3 Pooled testing

This is a problem about pooling—sometimes referred to as pool testing, pooled testing, or batch testing—means combining respiratory samples from several people and conducting one laboratory test on the combined pool of samples to detect a virus.

Suppose there are 2 individuals, each infected with probability $p \in [0, 1]$, independently. Assume that a test is absolutely accurate. That is probabilities of Type I and II errors are 0. In particular, if a sample contains a virus, then test always detects it. And, conversely, if there is no virus in a sample, then test is always negative. There are two types of testing procedures that one can use.

1. Individual testing. Each individual is tested separately.
2. Pooled testing. At stage 1 samples from both individuals are combined and tested. If test of stage 1 is negative, then both individuals are presumed negative. If test of stage 1 is positive, then we go to stage 2. At stage 2, individual 1 is tested. If test is negative, then clearly individual 2 is infected, and 1 is not. If test is positive, then we go to stage 3. At stage 3, individual 2 is tested. If test is negative, then individual 1 is infected and 2 is not. If test is positive, then both individuals are presumed infected.

Clearly, both procedures identify all infected individuals in all cases. However, the procedures might differ with respect to the required number of tests. Whereas individual testing always requires exactly 2 tests, pooled testing may require anywhere from 1 to 3 tests. We aim to choose a procedure to minimize the expected number of tests. The choice will depend on p . For instance, if p is close to 1, then pooling will most likely result in more tests than individual testing.

Determine the procedure which minimizes the expected number of tests for each value of p . Provide all the derivations and explanations.

3.4 Catching a train I

Alice is traveling from town A to town C with a stopover in town B. Here's the sequence of her journey. Alice boards Train 1 at town A heading to town B. Train 1 departs at 1 pm. Upon reaching town B, Alice needs to catch another train to town C. There are two available trains from B to C: Train 2, which departs at 2 pm, and Train 3, which departs at 3 pm. For each train i , the time it takes to reach its destination is represented by a random variable X_i which follows a uniform distribution $X_i \sim U(0.7, 1.1)$. The three random variables X_1, X_2, X_3 are independent and identically distributed. Assume that transferring from one train to another is instantaneous. However, if Alice misses her connecting train at town B, she has to wait for the next one. Let Y be a random variable representing the total duration (in hours) of Alice's journey from town A to town C, starting from 1 pm.

1. Find $P(Y \leq 2)$.
2. Find $\mathbb{E}(Y)$.

3.5 Catching a train II

Alice is traveling from town A_1 to town A_n with stopovers in towns $A_2, \dots, A_{n-1}$, where $n \geq 3$. The sequence of her journey is as follows:

- Alice boards Train 1 at town A_1 heading to town A_2 . Train 1 departs at 1 pm.
- Upon reaching town A_2 , Alice must catch another train to town A_3 , and so on.
- A train from town A_k to town A_{k+1} departs every hour on the hour: 1 pm, 2 pm, 3 pm, etc.
- For each train, the travel time to its destination is a random variable $X_i \sim \mathcal{U}(0.7, 1.1)$ hours.
- All X_i are independent and identically distributed.
- Transferring between trains is instantaneous.
- Alice always boards the first available train. If she misses the current one, she waits for the next hourly departure.

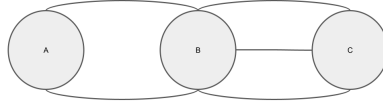
Let Y be the random variable representing the total duration (in hours) of Alice's journey from town A_1 to town A_n , starting from 1 pm.

1. Find $\mathbb{P}(Y \leq n - 1)$.
2. Find $\mathbb{E}(Y)$.
3. Compute $\text{Var}(Y)$.

3.6 Multiple choice exam

Alice prepares for a multiple-choice exam with probability $p \in (0, 1)$ and does not prepare with probability $1 - p$. If Alice studies, she answers each question correctly with probability $\frac{1}{2}$. If she does not study, she selects an answer randomly. Since each question has 4 answer choices (with only one correct), her probability of answering correctly in this case is $\frac{1}{4}$. The exam consists of 10 questions, and Alice answers each question independently.

1. What is the probability that Alice answers the first question correctly?
2. What is the probability that Alice answers all 10 questions correctly?
3. Let A_i be the event that Alice answers question i correctly. Are A_1 and A_2 independent? Answer *yes* or *no* and briefly explain your reasoning.
4. Suppose Alice answers all 10 questions correctly. Given this, what is the probability that she studied for the exam?



3.7 Subsets of a set

A set A has 10 elements. You randomly choose a subset of A (so that all subsets are equally likely to be chosen). Let the random variable X be the number of elements in the chosen subset.

1. Find $\mathbb{P}(X = 1)$.
2. Find $\mathbb{E}(X)$ (you must provide a number here).

3.8 Virus transmission

During a single contact a virus is transmitted from an infected to a healthy person with probability $p > 0$. Assume that the transmissions are independent across contacts.

1. What is the probability of getting infected after 2 independent contacts with infected individuals, given that each contact results in transmission with probability p ?
2. What is the probability of getting infected after 2 independent contacts, where each contact is with an individual who is infected with probability $q > 0$, independently, and transmission occurs with probability p if the contact individual is infected?
3. Suppose that after 2 independent contacts with individuals, each of whom is infected with probability $q > 0$, Alice is not infected. Given that Alice was not infected, what is the conditional probability that neither of Alice's contacts was infected?

3.9 Hiking with friends

Consider a network of hiking paths connecting three points as shown above. Alice and 10 strangers are planning to make a trip starting at point A, traveling through point B, and finishing at point C. At each junction, each traveler randomly and independently chooses a path to continue on. Every time Alice shares a path with a group of people, she becomes friends with everyone in that group. Assume that each person takes the same amount of time to traverse any given path.

1. Let X_1 and X_2 be random variables representing the number of friends Alice makes during the first and second legs of her hike, respectively. Are these random variables independent? Briefly explain.
2. What is the probability that X_1 is equal to 3?
3. What is the expected number of friends Alice will make by the end of her trip?
4. What is the probability that Alice becomes friends with everyone on the trip?

3.10 Table tennis scores

Alice and Bob are playing a table tennis game consisting of multiple rounds. Each round is independently won by Alice with probability p and by Bob with probability $1 - p$.

- Assume that the game ends as soon as one player reaches **11 points**.¹
 1. Find the probability that during the game, the score $(0, 1)$ (Alice: 0, Bob: 1) will appear at some point before the game ends.
 2. Find the probability that during the game, the score $(3, 2)$ (Alice: 3, Bob: 2) will appear at some point before the game ends.
 3. Find the probability that during the game, the score (x, y) will appear at some point before the game ends, where x and y are integers such that $0 \leq x, y \leq 10$.
 4. What is the probability that Alice wins the game?
- Assume that the game ends as soon as one player reaches **20 points**. Given a score (x, y) , let the corresponding last-digit score be (x', y') , where x' and y' are the last digits of x and y , i.e. if the score is $(11, 4)$ the last-digit score is $(1, 4)$.
 1. Find the probability that during the game, the last-digit score $(3, 2)$ will appear at some point before the game ends.
 2. Find the probability that during the game, the last-digit score (x, y) will appear at some point before the game ends, where x and y are integers such that $0 \leq x, y \leq 10$.
 3. What is the probability that Alice wins the game?

3.11 Miscellaneous problems

1. A corridor has 7 office doors, each occupied by some professor. There are no signs on office doors. A student is looking for a professor who occupies one of the offices but does not know which one. How many doors on average a student has to knock in before finding the right one (including the right one)? Assume that a student does not knock twice to the same door.
2. An urn has 3 white and 1 black ball. You take out 2 balls without replacement. What is the expected number of white balls in your hand?
3. You are organizing a round robin table tennis tournament (each player meets all other players in turn, each pair of players meets only once). Let X be a random variable representing the number of people who sign up for the tournament. Suppose you estimate that $\mathbb{E}[X] = 30$ and $\text{Var}[X] = 40$, however you do not know the distribution of X . What is the expected number of matches in the tournament?
4. The Rapid Test is used to determine whether someone has HIV (the virus that causes AIDS). The false-positive and false-negative rates are .027 and .080, respectively. A physician has just received the Rapid Test report that his patient tested positive. Before receiving the result, the physician assigned his patient to the low-risk group (defined on the basis of several variables) with only a 0.5% probability of having HIV. What is the probability that the patient actually has HIV?

¹According to the actual modern rules, a player must win by at least 2 points, so the actual number of point scored can be more than 11 (e.g., 11-9, 12-10).

5. The Pap smear is the standard test for cervical cancer. The false-positive rate is 0.636; the false-negative rate is 0.180. Family history and age are factors that must be considered when assigning a probability of cervical cancer. Suppose that, after obtaining a medical history, a physician determines that 2% of women of his patient's age and with similar family histories have cervical cancer. Determine the probability that the patient has cervical cancer given a positive Pap smear test.
6. You have a standard well-shuffled 52-card deck. On average, how many pairs of adjacent cards are there such that both cards are red? "Adjacent cards" means that cards lie next to each other in the deck. For example, at most, there are 25 pairs of adjacent red cards when all 26 red cards are one after another. Leave your answer as a fraction.
7. Suppose that a random variable X has the discrete uniform distribution on the integers 10, . . . , 20. Find the probability that X is even.
8. Suppose that two balanced dice are rolled, and let X denote the absolute value of the difference between the two numbers that appear. Determine and sketch the p.m.f. and c.d.f. of X .
9. In order to find out how many fish there are in a lake, 1000 fish are caught, marked and put back into the lake. For what number of fish in the lake will the probability be greatest of encountering 10 marked fish among 150 fish caught a second time?
10. How many ways are there to split 10 people into 2 teams with 3 and 7 people?
11. Find the probability that a randomly drawn 5-card hand will have 2 jacks, 2 queens, and a card of any other rank.
12. Laboratory mice are continuously exposed to radiation. Each month a mouse exposed to radiation develops certain cancer with a probability $p > 0$. Assume cancer occurrences each month are independent events. Suppose that on average mice develop cancer at the end of the 4th month. Find p .
13. Find the probability that a random binary sequence of length $2n$ will have a single 0.
14. A certain family has 7 children, consisting of 4 boys and 3 girls. Assuming that all birth orders are equally likely, what is the probability that the 3 eldest children are the 3 girls?
15. A couple makes babies, until it has at least one boy and at least one girl. Find the expected number of children this couple will have.
16. Consider the following lottery. You bet \$1 on one of the integers from 1 to 10 including. Then an integer is chosen uniformly at random. If your number is chosen then you win \$9, otherwise, you lose your \$1. Find the variance of your winnings.
17. An airline overbooks a flight, selling more tickets for the flight than there are seats on the plane (figuring that it's likely that some people won't show up). The plane has 100 seats, and 104 people have booked the flight. Each person will show up for the flight with probability 0.9, independently. Find the probability that there will be enough seats for everyone who shows up for the flight.
18. On average how many times you need to roll a die before all six different numbers have showed up? (Hint: use indicator random variables.)

19. In a certain city, three newspapers A, B, and C are published. Suppose that 60 percent of the families in the city subscribe to newspaper A, 40 percent of the families subscribe to newspaper B, and 30 percent subscribe to newspaper C. Suppose also that 20 percent of the families subscribe to both A and B, 10 percent subscribe to both A and C, 20 percent subscribe to both B and C, and 5 percent subscribe to all three newspapers A, B, and C. What percentage of the families in the city subscribe to at least one of the three newspapers?
20. Each of 10 balls is independently placed into one of 10 boxes, with all boxes equally likely. What is the probability that exactly one box is empty?
21. A fair coin is tossed 64 times. What is the distribution of the number of heads? Name the distribution and specify its parameters.
22. A club consists of 8 seniors, 10 juniors, and 12 sophomores. An organizing committee of size 4 is chosen randomly (with all subsets of size 4 equally likely). Find the probability that the committee has at least one representative from each of the senior, junior, and sophomore classes.
23. The screens used for a certain type of cell phone are manufactured by 3 companies, A, B, and C. The proportions of screens supplied by A, B, and C are 0.4, 0.3, and 0.3, respectively, and their screens are defective with probabilities 0.01, 0.02, and 0.03, respectively. Given that the screen on such a phone is defective, what is the probability that Company A manufactured it?
24. Suppose events A and B are disjoint. Are they independent? Say yes or no and provide a brief explanation.
25. In a chess tournament, n games are being played, independently. Each game ends in a win for one player with probability 0.3 and ends in a draw (tie) with probability 0.7. Let X be the number of players whose games end in draws.
 - Find support of X .
 - Find PMF of X . Note you have to specify a probability for each number on the support.
26. Alice and Bob each randomly and independently choose 3 out of 5 different classes. Let X be the number of classes taken by both students at the same time. Find $\mathbb{E}(X)$.

4 Continuous distributions

4.1 Basic distributions I

Let a random variable X have the following probability density function:

$$f(x) = \begin{cases} ce^{x-1} & \text{if } x \in [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

1. Find a constant $c \in \mathbb{R}$ such that f is a valid probability density function.
2. Find the value of the cumulative distribution function (cdf) of X evaluated at a point $x \in (0, 1)$ (you must provide a function of x).

4.2 Basic distributions II

Suppose X is a random variable uniformly distributed on the interval $[-1, 1]$.

1. What is the expectation of X ?
2. What is the variance of X ?
3. What is the probability that $X^2 \leq 0.25$?
4. What is the expectation of $\max\{0, X\}$?

4.3 Basic distributions III

A random variable X has the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & \text{if } x < 0, \\ 2x, & \text{if } 0 \leq x \leq \frac{1}{3}, \\ 0.5(x + 1), & \text{if } \frac{1}{3} < x \leq 1, \\ 1, & \text{if } x > 1. \end{cases}$$

1. Find $\mathbb{P}\left(\frac{1}{6} \leq X \leq \frac{3}{4}\right)$.
2. Find the value of the probability density function of X at $1/2$, that is, find $f(1/2)$.
3. Find $\mathbb{E}(X)$.

4.4 Basic distributions IV

There are 100 students in a statistics class. Let X_i be a random variable representing the fraction of a working day that student i spends studying for the final exam. Suppose $X_1, X_2, \dots, X_{100}$ are independent random variables with probability density function (pdf):

$$f_X(x) = \begin{cases} 6x(1 - x), & \text{if } x \in (0, 1), \\ 0, & \text{otherwise.} \end{cases}$$

1. What is the probability that student i spends **less than one third** of her day studying for the exam?
2. What is the **expected fraction** of a day that student i spends studying for the exam?
3. Suppose student i **passes the exam** if and only if she devotes **more than one third** of the day to studying. What is the **expected number of students** who pass the exam?

4.5 Dating normally

Suppose the distribution of student height at NU is approximately normal. In particular, the height of girls follows a normal distribution with mean 165 cm and variance 100. The height of boys follows a normal distribution with mean 175 cm and variance 100.

1. What is the probability that a randomly chosen girl is taller than 150 cm?

2. Suppose a girl and a boy can date if and only if the boy is taller. Assem is a girl whose height is 180 cm. There are 2000 boys at NU. On average, how many of them are taller than Assem (and thus could date her)?
3. Pedro is currently dating Assem from the previous question. What is the probability that Pedro is taller than 190 cm?

4.6 Artillery precision

M982 Excalibur is a modern 155 mm artillery round that uses GPS guidance to steer itself towards its target. Suppose that a continuous random variable ε measures the distance (absolute deviation) of the shot from the target in meters and has pdf $f(\varepsilon) = 0.02(10 - \varepsilon)$ if $\varepsilon \in [0, 10]$ and $f(\varepsilon) = 0$ otherwise. When multiple shots are fired, the deviations are independent. Suppose that to destroy a given target it is necessary and sufficient to strike at least once within 3 meters from it.

1. What is the probability that a target will be destroyed with a single round?
2. Let X be a discrete random variable representing the number of Excalibur rounds required to destroy a single target. Find the distribution of X .
3. Let X be a discrete random variable representing the number of Excalibur rounds required to destroy a single target. Find $\mathbb{E}[X]$.
4. Each of 10 targets is fired at once. What is the probability that exactly 5 targets will be destroyed?
5. Circular Error Probable (CEP) is a statistical measure that is commonly used to describe the accuracy of weapons, including artillery rounds. CEP is defined as the radius of a circle within which 50% of the shots fired are expected to fall. In other words, if a weapon is fired multiple times at the same target, with the same conditions and aim point, the CEP represents the circle within which half of the shots will land. Calculate CEP of an Excalibur round given the distribution of deviations ε .

4.7 Waiting for a bus

Alice is waiting for a bus at a bus stop. She needs to take a bus number 10 or 12, and she takes the first suitable bus that arrives. The arrival time of bus 10 is exponentially distributed with rate $\lambda_{10} = 10$, and the arrival time of bus 12 is exponential with rate $\lambda_{12} = 5$. Moreover, arrival times of buses 10 and 12 are independent.

1. What is the probability that Alice will take bus number 10 instead of bus number 12?
2. Let X be the amount of time Alice has to wait. Then the probability density function of X is given by $f_X(x) =$ (Do not forget to specify the support of X)
3. How long on average does Alice have to wait for a bus?

4.8 Waiting for a kidney

Alice is at the top of the waiting list for a deceased donor kidney. It is a good approximation to assume that compatible kidneys arrive randomly, with the waiting time t (in years) exponentially distributed with rate $\lambda = 0.5$.

1. What is the probability that Alice will have to wait exactly one year?
2. What is the probability that Alice will have to wait less than two years?
3. Now suppose that Alice is in the k^{th} place on the waiting list, where the list is composed of people with the same blood type. Then she can receive a kidney only after each of the $k - 1$ people ahead of her on the list receives a kidney. How long, on average, does she have to wait for a transplant? Provide a number.
4. Given the conditions from the previous question, what is the variance of the time she has to wait to receive a kidney? Provide a number.
5. Suppose that Alice enters the list randomly, so with equal chance she can be in any position from 1st to k^{th} . How long, on average, does she have to wait for a transplant? Provide an expression in terms of k .
6. Suppose Alice turns to a black market, asking a local mafia to find a donor with a compatible kidney for sale. It also takes time for the mafia to find a willing (or unwilling) donor. This waiting time t is exponentially distributed, but with a higher rate $\lambda = 1$. Alice is at the top of the legal waiting list and gets a transplant as soon as either a legal kidney becomes available or the mafia finds a compatible kidney. How long, on average, does she have to wait for a transplant? Provide a number.

4.9 Miscellaneous problems

1. The amount of time it takes for a student to complete a statistics quiz is uniformly distributed between 30 and 90 minutes. Find the probability that the student completes the quiz in a time between 30 and 50 minutes.
2. The manager of a gas station has observed that the times required by drivers to fill their car's tank and pay are quite variable. In fact, the times are exponentially distributed with a mean of 8 minutes. What is the probability that a car can complete the transaction in less than 8 minutes?
3. The Tesla Model S85D is an electric car that the manufacturer claims can travel 270 miles on a single charge. However, the actual distance depends on a number of factors including speed and whether the car is driven in the city or on highways. Suppose that the distance is a normally distributed random variable with a mean of 200 miles and a standard deviation of 20 miles. An owner of this model intends to travel to a nearby city and return on the same charge. If the total distance is 210 miles, what is the probability that car makes it without running out of power?
4. Suppose that X is a normally distributed random variable with mean 10 and variance 2. What is the probability that X is below 10?
5. Alice has taken a PCR test and is waiting for the result. She has to take a plane in exactly 3 days from now, and she can fly only if she has a negative test by that time. Alice is infected with probability 0.5. The waiting time for the result is exponential with $\lambda = 2$, and is independent of Alice being infected or not. Assuming that test is perfectly accurate, what is the probability that she will be able to fly as planned?
6. Suppose U is a random variable uniform on $[0, 9]$. What is the probability that a square with a side length equal to U has an area less than 9?

7. A continuous random variable X is uniformly distributed on $[-1,1]$. What is the probability that the absolute value of X exceeds 0.8?
8. Z is a standard Normal random variable and $W = 3Z + 2$. Find the variance of W .
9. Z is a standard Normal random variable and $W = 3Z + 2$. What is the distribution of W ? Name the distribution and specify its parameters.
10. Let $X \sim \text{Bern}(3/4)$. Then:
 $\mathbb{E}(X) =$
 $\text{Var}(X) =$
11. A circle with a random radius R is generated, where R is a random variable that takes positive values. The expectation of R is 10, and the standard deviation is 4. Find the expected area of the circle.
12. (2 points) A fair coin is tossed 64 times. Every time head comes up you get €3, and if tails comes up you pay €1. How much should one expect to earn from playing this game?
13. (3 points) A fair coin is tossed 64 times. Every time head comes up you get €3, and if tails comes up you pay €1. Use CLT to estimate the probability that earnings from this game are below €72.²
14. Determine the least squares line (its slope b_1 and the intercept b_0) that fits the following data: (0,0), (0,1), (1,1), where the first coordinate is x and the second is y .
 - $b_1 =$
 - $b_0 =$
15. Let X and Y be i.i.d. r.v.s uniform on $[0,1]$ and let $M = \max\{X, Y\}$. Find variance of M .

5 Applications

5.1 Posted-price mechanism I

Pedro is selling his car using a posted-price mechanism. There are two potential buyers, Alice and Bob. Alice's and Bob's values are independent random variables given by, correspondingly,

$$v_A \sim U[0, 3], v_B \sim U[0, 1].$$

Buyers know their own values, while the seller does not. Pedro posts a price $p \geq 0$, and then sells to a buyer who wants to buy given this price (if more than a single buyer wants to buy then one of them is randomly chosen). A buyer $i \in \{A, B\}$ wants to buy if $p \leq v_i$.

1. (1 point) What is the probability that Alice's value is higher than 1?
2. (2 points) Find the variance of Alice's value.
3. (2 points) What is the probability that Bob's value is higher than Alice's value?

²Use the attached Standard Normal table.

4. (3 points) Given that Bob's value is higher than Alice's, what is the probability that Alice's value is less than 0.5?
5. (3 points) Let $M = \min\{v_A, v_B\}$. Find $P(M < 0.5)$.
6. (4 points) Let $M = \min\{v_A, v_B\}$. Find $\mathbb{E}(M)$.
7. (2 points) Suppose Pedro sets $p = 0.5$. What is the probability that at least one of the buyers will want to buy?
8. (2 points) Suppose Pedro sets $p = 0.5$. What is his expected revenue?
9. (4 points) Find price p that maximizes Pedro's expected revenue.

5.2 Posted-price mechanism II

Alice is selling her car using a posted-price mechanism. There are n potential buyers. Buyers have valuations iid with continuous uniform distribution on $[5, 10]$. Buyers know their own valuations, while Alice does not. Alice posts a price, and then each buyer decides to buy or not given this price (if more than a single buyer wants to buy then one of them is randomly chosen). Find a price (in terms of n) that maximizes the expected revenue of Alice.

5.3 Posted-price mechanism III

Alice is selling her car using a posted-price mechanism. There are 3 potential buyers. Buyers have valuations that are independently and identically distributed with the following probability density function:

$$f(x) = \begin{cases} 3x^2 & \text{if } x \in [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

Buyers know their own valuations, while Alice does not. Alice posts a price, and then each buyer decides whether to buy or not, given this price. If more than one buyer wants to buy, one of them is randomly chosen.

Find a price that maximizes the expected revenue of Alice.

5.4 Monopolist with Uncertain Demand

A monopolist operates in a market with uncertain demand. The demand function is given by:

$$q(p) = \max\{1 - \theta p, 0\},$$

where p is the price, and θ is a random variable representing a demand shock. The CDF of θ is:

$$F(\theta) = \begin{cases} 0, & \text{if } \theta \leq 0, \\ \theta^3, & \text{if } 0 < \theta < 1, \\ 1, & \text{if } \theta \geq 1. \end{cases}$$

- (a) (3 points) Find the median value of the demand shock θ .
- (b) (3 points) Compute the expected value of the demand shock θ .
- (c) (3 points) Find the probability that the revenue is strictly positive when $p = \frac{3}{2}$.

- (d) (3 points) Find the price p that minimizes the variance of the seller's revenue.
- (e) (4 points) Calculate the expected revenue when $p = \frac{3}{2}$.
- (f) (5 points) Derive the expected revenue-maximizing price.

5.5 Revenue in selling mechanisms

Alice is selling her car. Bob and Charlie are potential buyers. Their valuations, v_B and v_C , are independent random variables drawn from a uniform distribution on $[0, 2]$. Each buyer knows their own valuation, but Alice does not know them. She is considering three different mechanisms to sell her car.

Answer exactly one of the three questions below; state which question you answer.

1. Posted Price Mechanism.

Alice posts a single price p . Each buyer independently decides whether to buy at that price. If both buyers want to buy, one of them is chosen at random to purchase the car.

- (a) (4 points) Express Alice's expected revenue as a function of p .
- (b) (2 points) Find the value of p that maximizes Alice's expected revenue.

2. First-Price Sealed-Bid Auction.

Each buyer i submits a bid b_i . Suppose each buyer bids $b_i = \frac{v_i}{2}$. Alice sells the car to the highest bidder, who pays the amount of their own bid.

- (a) (3 points) Let X be the random variable denoting Alice's revenue. Compute $\Pr(X \leq 1/3)$.
- (b) (3 points) Compute $\mathbb{E}[X]$.

3. Second-Price Sealed-Bid Auction.

Each buyer i submits a bid b_i . Suppose each buyer bids truthfully, i.e., $b_i = v_i$. Alice sells the car to the highest bidder, who then pays the second-highest bid.

- (a) (3 points) Let Y be the random variable denoting Alice's revenue. Compute $\Pr(Y \leq 1/3)$.
- (b) (3 points) Compute $\mathbb{E}[Y]$.

5.6 Newsvendor model I

A coffee shop owner orders fresh milk and pastry every day. Let $s_m \geq 0$ be the amount of milk ordered (in liters), and $s_p \geq 0$ be the amount of pastry ordered (in kilograms). The daily demand for milk (in liters) is a random variable $X \sim U[0, 10]$. The daily demand for pastry (in kilograms) is a random variable Y with the pdf:

$$f(y) = \begin{cases} 0, & \text{if } y \notin [0, 1], \\ 6y, & \text{if } y \in [0, \frac{1}{3}), \\ 3(1 - y), & \text{if } y \in [\frac{1}{3}, 1]. \end{cases}$$

Both X and Y are independent. The wholesale price for milk is \$1 per liter, and the shop sells milk at \$5 per liter (so the profit per liter sold is \$4). The wholesale price for pastry is \$5 per kilogram, and the shop sells it at \$6 per kilogram (so the profit per kilogram sold is \$1).

1. (3 points) Find the expected daily demand for **pastry**.
2. (3 points) Find the variance of the daily demand for **milk**.
3. (2 points) Suppose the coffee shop orders 10 liters of milk and 1 kilogram of pastry. What is the total expected profit?
4. (2 points) Suppose the coffee shop orders $\frac{1}{2}$ kilogram of pastry. Find the probability that it runs out of pastry during the day.
5. (3 points) Suppose the coffee shop orders 5 liters of milk. What is the expected profit from milk sales?
6. (4 points) Determine the amount of milk the shop should order to maximize its expected profit from milk sales.
7. (3 points) Assume the shop owner orders the amounts of milk and pastry that maximize total expected profit. Which of the two (milk or pastry) is more likely to run out during the day? Briefly explain (no detailed calculations needed).

5.7 Newsvendor model II

Chaplin movie theater has to schedule the premiere showing of a new movie. They decide to screen the movie in two versions, original and dubbed. Let X be a random variable representing the number of viewers who come to see the dubbed version, and Y be a random variable representing the number of viewers who come to see the original version. For simplicity, assume that X is a continuous random variable uniformly distributed on $(0, 300)$; Y is a continuous random variable uniformly distributed on $(0, 200)$; and X and Y are independent. So, the dubbed version is more popular. Assume that viewers who came to see a particular version are not going to watch the other version if their preferred choice is not available.

- Suppose the cinema allocates 200 seats to the dubbed version, and 100 seats to the original.
 1. What is the expected number of viewers who come to see the movie (any version)?
 2. What is the expected number of tickets sold for the original version and for the dubbed version? Why is this number different from the previous answer?
 3. Let $Z = \max\{X, Y\}$.
 - (a) Find CDF of Z .
 - (b) Find PDF OF Z .
 - (c) Find $E[Z]$.
- Suppose now that the cinema can decide how to allocate its 300 seats across the two movies. Also let the ticket price be the same for the both movies. Find the allocation of seats that maximizes the expected revenue of the cinema. For simplicity assume that any continuous allocation is possible, that is the cinema chooses $x \in [0, 300]$ to allocate to the original, and $300-x$ to the dubbed version.

5.8 Newsvendor model III

A gas station sells gas for \$1 per liter and buys it using the wholesale price of \$0.5 per liter. A storage capacity of a gas station is 700 liters per day, that is it can store any quantity of gasoline up to this capacity. A gas station cannot sell more gasoline per day than it currently stores. Suppose the demand for gasoline is uniformly distributed between 0 and 1500. Profit of a gas station is given by its revenue from selling gasoline minus the cost of buying it from a wholesaler. Provide a numerical answer to each question.

5.9 Newsvendor model IV

A fish restaurant sells fish for a fixed price of \$50 per kg and buys it from a supplier for a fixed cost of \$10 per kg. All fish not sold during the day goes bad. A restaurant manager has estimated that daily demand for fish in kg is uniformly distributed on $[0, 40]$. Find a profit-maximizing quantity of fish that the manger should purchase from the supplier.

1. Find the expected daily profit of the gas station when it decides to store 500 liters of gasoline.
2. Given the current storage capacity, find the quantity of gasoline the gas station must store in order to maximize its expected daily profit.

5.10 Supermarket Revenue Analysis

A supermarket in block A1 sells three brands of pasta: Barilla (B), Garofalo (G), and Sultan (S), with prices:

Barilla: 700 tenge, Garofalo: 2000 tenge, Sultan: 300 tenge.

The number of customers per day follows $X \sim \text{Bin}(100, 0.5)$. Each customer independently chooses what to do. Specifically, a customer can buy:

- Barilla with probability 0.2, or
 - Garofalo with probability 0.1, or
 - Sultan with probability 0.3, or
 - no pasta with probability 0.4.
- (a) Compute the expected number of customers per day.
 - (b) Write an arithmetic expression for the probability that the number of customers is ≤ 98 . Do not compute the value.
 - (c) Find the expected number of packs of Barilla sold per day.
 - (d) Compute the daily expected revenue from selling pasta.
 - (e) Write an arithmetic expression for the variance of daily revenue. Do not compute the value.
 - (f) Using the Central Limit Theorem, find the probability that daily revenue exceeds 29,000 tenge if the standard deviation is assumed to be known to be 5,000.

6 Joint distributions

6.1 Portfolio choice I

Alice is deciding how to distribute her wealth between two available assets. Let X and Y be random variables representing the returns of the first and second assets, respectively. Let $\alpha \in [0, 1]$ be the share of the first asset in Alice's portfolio, and $1 - \alpha$ be the share of the second asset. The joint CDF of X and Y is given in the table below:

$X \backslash Y$	$Y = 0$	$Y = 1$
$X = 0$	$1/6$	$1/2$
$X = 1$	$1/2$	1

- (a) (3 points) Find the probability that $XY > 0$.
- (b) (3 points) Given $X = 1$, find the probability that $Y = 1$.
- (c) (3 points) Compute $\mathbb{E}(X)$.
- (d) (4 points) Calculate $\text{Cov}(X, Y)$.
- (e) (4 points) If Alice's utility function is $U(\sigma_\alpha^2) = -\sigma_\alpha^2$, where σ_α^2 is the variance of portfolio returns, determine the optimal α .

6.2 Portfolio choice II

Consider two assets, whose returns are represented by random variables X and Y , respectively. Their joint distribution is provided by the following table:

	$Y = 0$	$Y = 10$	$Y = 100$
$X = 0$	0.1	0.1	0
$X = 10$	0	0.3	0.5

- 1. (3 points) Are X and Y independent? Explain.
- 2. (3 points) Find $P(X = 10)$.
- 3. (3 points) Find $P(X \cdot Y \geq 100)$.
- 4. (4 points) Find $E(XY)$.
- 5. (4 points) Given that $Y > 0$, what is the probability that $X > 0$?
- 6. (4 points) Find $\text{Var}(Y)$.
- 7. (5 points) Find $\text{Cov}(X, Y)$.
- 8. (3 points) Find $\text{Corr}(X, Y)$.
- 9. (3 points) Let $\alpha \in [0, 1]$ represent the share of the first asset in a portfolio. Determine the value of α that maximizes the portfolio's return.
- 10. (5 points) Find the value of α that minimizes the portfolio return variance.

	$Y = 0$	$Y = 10$
$X = 0$	0.5	a
$X = 15$	$0.5 - a$	0

6.3 Portfolio choice III

Consider two assets, whose returns are represented by random variables X and Y , respectively. Their joint distribution is provided by the following table:

Provide complete solutions and numerical answers to the following questions.

1. Assume $a = 0.3$.

- Find the probability that $X = 0$.
- Find the probability that $X + Y \leq 10$.
- Find the probability that $X < Y$.
- Are X and Y independent? Provide proof.
- Compute the expected values $E[X]$ and $E[Y]$.
- Compute the variances $\text{Var}(X)$ and $\text{Var}(Y)$.
- Given that $Y = 0$, find $P(X = 0 \mid Y = 0)$.
- Compute the covariance and correlation between X and Y .
- Let $\alpha \in [0, 1]$ represent the share of the first asset in a portfolio. Determine the value of α that maximizes the portfolio's return.
- Plot the portfolio return variance as a function of α .
- Find the value of α that minimizes the portfolio return variance.
- Suppose an investor has the utility function:

$$U(\mu, \sigma^2) = \mu - \lambda \sigma^2,$$

where μ is the expected portfolio return, σ^2 is the variance, and $\lambda > 0$ indicates risk aversion. Identify the portfolio the investor will select.

2. Now assume you can choose $a \in [0, 0.5]$.

- Find the value of a that allows you to construct a portfolio with the highest possible expected return.
- Find the value of a that allows you to construct a portfolio with the lowest variance.

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.25	0.25	0
$Y = 1$	0	0.25	0.25

6.4 Portfolio choice IV

There are two assets. Let X and Y be random variables equal to the return on the first asset and the second asset, respectively. The joint distribution of X and Y is given by the table below.

Provide a numerical answer to each question.

1. Find the probability that X is equal to 1.
2. Are X and Y independent?
3. Suppose you know that Y is 0. Find the probability that X is also 0.
4. Find the expected value of X .
5. Find the expected value of $|X - Y|$.
6. Find the variance of X .
7. Find the covariance between X and Y .
8. Let $\alpha \in [0, 1]$ be the share of the first asset in a portfolio.
 - (a) Find α that maximizes the expected return of the portfolio.
 - (b) Find α that minimizes the variance of return of the portfolio.

6.5 Risk and Revenue in Selling Mechanisms

Alice plans to sell her only concert ticket to one of her classmates, Bob or Carol. Both students may value the ticket differently, and Alice is unsure how much each is willing to pay. Their valuations are independent and uniformly distributed:

$$V_B, V_C \sim U[0, 2].$$

Alice herself has no value for the ticket.

Different selling mechanisms can generate different average revenues for the seller. However, Alice may also care about how *risky* her revenue is. If she urgently needs money for rent tomorrow, a smaller but more predictable revenue may be preferable to a riskier one with a higher mean. To study this, we compare the expected revenue and its variance under three mechanisms.

Second-Price Auction (SPA)

Bob and Carol each submit sealed bids equal to their valuations. The ticket goes to the higher bidder, who pays the second-highest bid.

- (a) Express the seller's revenue R_{SPA} in terms of V_B and V_C .
- (b) Compute $E[R_{SPA}]$ and $Var(R_{SPA})$.

First-Price Auction (FPA)

Each buyer bids $b_i = \frac{1}{2}v_i$. The highest bidder wins and pays their own bid.

- (a) Express the seller's revenue R_{FPA} in terms of V_B and V_C .
- (b) Derive $E[R_{FPA}]$ and $Var(R_{FPA})$.

Posted Price Mechanism (PPM)

Alice posts a price $p \in [0, 2]$. Each buyer decides whether to buy at that price. If both are willing to buy, one of them is chosen uniformly at random to purchase the ticket at price p ; otherwise, no sale occurs.

- (a) Derive the probability of sale $\Pr(\text{sale at } p)$ as a function of p .
- (b) Express the seller's expected revenue $E[R_{PP}(p)]$ as a function of p .
- (c) Find the price p^* that maximizes expected revenue and compute $E[R_{PP}(p^*)]$.
- (d) Derive $Var(R_{PP}(p))$ as a function of p and evaluate it at p^* .
- (e) Sketch $Var(R_{PP}(p))$ as a function of p . Explain intuitively why the variance approaches zero for very small or very large p .
- (f) Differentiate $Var(R_{PP}(p))$ to find the value of p that maximizes the variance. Explain why this price differs from p^* .
- (g) Plot $E[R_{PP}(p)]$ on the vertical axis and $Var(R_{PP}(p))$ on the horizontal axis for all $p \in [0, 2]$. This curve represents all possible combinations of mean and variance Alice can achieve by choosing different posted prices.
 - Can some prices be ruled out as *dominated* (yielding both lower mean and higher variance than another price)?
 - Graphically identify the *efficient set* of prices—those not dominated by any others.
 - Compare this set to the concept of the *efficient frontier* in portfolio theory.

Alice's Mean–Variance Preferences

Suppose Alice is risk-averse and evaluates mechanisms according to the mean–variance utility function:

$$U = E[R] - \frac{\lambda}{2}Var(R), \quad \lambda \geq 0,$$

where λ measures her degree of risk aversion. A risk-neutral Alice has $\lambda = 0$; larger values of λ indicate a stronger dislike of variability.

- (a) Suppose $\lambda > 0$. Compute U_{SPA} and U_{FPA} . Which mechanism does Alice prefer, and why?
- (b) Using your earlier results, write Alice's utility as a function of the posted price:

$$U(p) = E[R_{PP}(p)] - \frac{\lambda}{2}Var(R_{PP}(p)).$$

- (c) For each $\lambda \geq 0$, determine the utility-maximizing price $p(\lambda)$. Explain intuitively why a risk-averse seller might choose a price lower than the expected revenue-maximizing one.
- (d) Depict all attainable mean-variance pairs for posted prices (the frontier) together with the two auction points (SPA and FPA). On the same graph, draw several indifference curves for different values of λ . Indicate which mechanism (and which price) Alice would choose for low, moderate, and high risk aversion.

6.6 Continuous join distribution I

Consider random variables X and Y with joint density:

$$f_{X,Y}(x,y) = \begin{cases} c, & \text{if } x \geq 0, y \geq 0, 2x + y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Provide a complete solution and numerical answers for:

1. Find constant c .
2. Find the marginal distributions of X and Y .
3. Find the probability density function of X given $Y = \frac{1}{2}$.
4. Find the expected values of X and Y .
5. Find the variances $\text{Var}(X)$ and $\text{Var}(Y)$.
6. Compute $\text{Cov}(X, Y)$ and $\text{Corr}(X, Y)$.

6.7 Continuous join distribution II

A pharmaceutical company is evaluating an R&D project. The value of the successful project, V , is uniformly distributed on $[0, 1]$, and the difficulty η (independent of V) is uniformly distributed on $[0, 2]$. The R&D cost is η^2 .

1. Find the probability that the project results in negative profit.
2. Compute the expected profit from the project.

6.8 Discrete joint distributions

Alice and Bob are taking an exam consisting of two questions. Each question is independently classified as either *hard* with probability p or *easy* with probability $1 - p$. Alice and Bob each answer a *hard* question correctly with probability q_H and an *easy* question correctly with probability q_E , where $q_H < q_E$. Define event A_i as Alice answers question i correctly, and event B_i as Bob answers question i correctly. Assume that A_i and B_i are *conditionally independent* given the difficulty level of question i , and that A_i and B_j are independent for all $i \neq j$.

1. Intuitively explain why events A_i and B_i are not independent.
2. Find the conditional probabilities $P(A_i | B_i)$ and $P(B_i | A_i)$.
3. Provide a formal argument demonstrating that events A_i and B_i are not independent.

4. Let X denote the number of questions Alice answers correctly, and let Y denote the number of questions Bob answers correctly. Do you think X and Y are positively, negatively correlated or independent? Explain your reasoning intuitively.
5. Construct the joint probability mass function (pmf) of X and Y .
6. Derive the marginal pmfs of X and Y .
7. Formally check your conjecture about X and Y in question 4 (you need to construct a product table).
8. Compute the covariance and correlation between X and Y .
9. Given that Alice answered both questions correctly, calculate the probability that Bob also answered both questions correctly.

6.9 Random meeting

Alice and Bob arrange to meet for lunch on a certain day at noon. However, neither is known for punctuality. They both arrive independently at uniformly distributed times between noon and 1 pm on that day. Each is willing to wait up to 20 minutes for the other to show up.

1. (3 points) What is the probability that Alice arrives first?
2. (3 points) What is the probability that Alice arrives before 12.45 pm?
3. (3 points) What is the probability that Alice and Bob both arrive before 12.45 pm?
4. (3 points) Let X_A be a random variable representing Alice's arrival time and X_B be a random variable representing Bob's arrival time. Find $Cov(X_A, X_B)$.
5. (4 points) When does the person who arrives second arrives in expectation? (Write the expected arrival time.)
6. (5 points) What is the probability they will meet for lunch that day?
7. (6 points) Given that Alice has arrived after 12.45 pm, what is the probability that they met for lunch that day?

7 Statistics

7.1 Nose picking and randomized response

Suppose we are conducting a survey to estimate the proportion of students who regularly pick their nose. Let p denote this proportion in the population. The survey asks only whether a respondent does or does not pick their nose.

1. Suppose $p = 0.5$. Using the Central Limit Theorem, estimate the probability that in a random sample of 20 people, at least 12 answer "yes."
2. Let p be unknown and n the number of people surveyed. Let X_n be the sample proportion of respondents who answer "yes." Using the Central Limit Theorem, estimate the smallest sample size n required to ensure that X_n is within 0.01 of the true value of p with probability at least 0.9.

3. Now suppose respondents may not tell the truth. To mitigate this, each participant is instructed to secretly flip a fair coin before answering. If the coin lands heads, they answer truthfully; if tails, they answer “yes” regardless of the truth. This way, an individual’s response does not directly reveal their behavior, reducing the incentive to lie. Suppose k out of 20 people answer “yes” under this mechanism. Provide an unbiased estimate of p .

7.2 Exam question: statistics I

This course is worth 6 ECTS credits, which implies a total workload of at least 150 hours. Of these, 35 hours are allocated to lectures and 2 hours to the exams, leaving 113 hours for independent study and problem sets, assuming full attendance. Spread over 14 weeks, this corresponds to approximately 8 hours per week.

To evaluate whether the actual workload matches this expectation, I surveyed 16 randomly selected students, asking them to estimate how many hours per week they spend studying for this course, not including lectures.

- Sample mean: $\bar{x} = 7$ hours
- Sample standard deviation: $s = 4$ hours

We want to test whether students study significantly less than the expected 8 hours per week. Formally, we test the hypotheses:

$$\begin{aligned} H_0 : \mu &\geq 8 && \text{(Students study at least 8 hours per week on average)} \\ H_1 : \mu &< 8 && \text{(Students study less than 8 hours per week)} \end{aligned}$$

1. (2 points) Compute the value of t-statistic from the survey data.
2. At the 1% significance level, is there sufficient evidence to reject H_0 ? Provide a corresponding rejection region for the value of t-statistic and conclude.³
 - (1 point) Rejection region for t-statistic:
 - (1 point) Reject or Not
3. At the 10% significance level, is there sufficient evidence to reject H_0 ? Provide a corresponding rejection region for the value of t-statistic and conclude.
 - (1 point) Rejection region for t-statistic:
 - (1 point) Reject or Not
4. (2 points) What does it mean to **not reject** H_0 at 10% significance? Choose one.
 - There is more than a 10% chance that H_0 is true.
 - We have proven that H_0 is true with 90% probability.
 - p -value is less than 0.10, providing evidence against H_1 .
 - **p -value is more than 0.10, not providing sufficient evidence against H_0 .**

³Use the attached t-distribution table.

5. (3 points) Calculate a 90% confidence interval for the average number of study hours.
6. (2 points) What does it mean that a number x does not belong to 90% confidence interval? Choose one.

- If x was a true average, then obtaining the sample mean as extreme or more extreme as ours has less than 10% probability.
- If x was a true average, then obtaining the sample mean as extreme or more extreme as ours has more than 10% probability.
- x is not an unbiased estimator of true average with 90% probability.
- x is the true average with 10% probability.

7.3 Courier Service's Claim

A courier service claims that its average delivery time for local deliveries is less than 20 minutes. To test this claim, consider the following hypotheses:

$H_0 : \mu \geq 20$ (The average delivery time is at least 20 minutes.)

$H_1 : \mu < 20$ (The average delivery time is less than 20 minutes, supporting the courier's claim.)

A random sample of 16 delivery times to an address across town was recorded:

- Sample mean: $\bar{x} = 24$ minutes,
 - Sample standard deviation: $s = 8$ minutes.
1. At the 1% significance level, is there sufficient evidence to support the courier service's claim? Compute the test statistic and determine whether to reject H_0 . Clearly state your conclusion.
 2. Explain the exact mathematical meaning of your conclusion. What does it mean to **reject** or **fail to reject** H_0 regarding the courier service's claim?
 3. Calculate a 95% confidence interval for the mean delivery time based on the sample data.
 4. Explain the mathematical meaning of the confidence interval. How does it relate to the true mean delivery time?
 5. Intuitively, what conclusion can be drawn about the courier service's claim based on the sample data?

7.4 Miscellaneous problems

1. A certain town is served by two hospitals. In the larger hospital, about 45 babies are born each day. In the smaller one, about 15 babies are born each day. Although the overall proportion of girls is about 50%, the actual proportion at either hospital may be greater or less on any day. At the end of a year, which hospital is more likely to have the greater number of days on which more than 60% of the babies born were girls?
2. You flip a fair coin 100 times, and then compute a fraction of Heads. Use the central limit theorem to find the probability that this fraction is between 0.4 and 0.6?

3. Consider 25 independent draws from the continuous uniform distribution on $[0,1]$. Using the central limit theorem find the probability that the average of the draws is below 0.4.
4. A business student claims that, on average, an MBA student is required to prepare more than five cases per week. To examine the claim, a statistics professor asks a random sample of 10 MBA students to report the number of cases they prepare weekly. The results are exhibited here. Can the professor conclude at the 5% significance level that the claim is true, assuming that the number of cases is normally distributed with a standard deviation of 1.5? Data: 2, 7, 4, 8, 9, 5, 11, 3, 7, 4